Improved high-precision UAV small object detection model based on YOLOv8

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Abstract

Object detection for unmanned aerial vehicle (UAV) has always been a popular research direction. However, the detection often involves the issue of objects with small pixel values. In response to this issue, this paper hopes to have a detection model that can achieve precise detection of UAV small objects, while also requiring the characteristic of occupying fewer resources. Therefore, this paper proposed an improved model based on YOLOv8 [1], high-precision YOLOv8, which has good performance in UAV small object detection and has been optimized for the balance between performance and resources. Firstly, this paper made improvements to the backbone network of YOLOv8 and proposed a new backbone network; In addition, for the detection of UAV small objects, this paper designed new feature fusion modules and a new detection head. Finally, this paper used MPDIoU [2] as the bounding box loss, adopting the shortest point distance method to compare any two bounding boxes, considering more computational factors and avoiding the failure of general intersection over union (IoU). At the same time, this paper used the LAMP [3] pruning method to simplify the model, reduce computational complexity, and thus improve the model's localization ability. On the VisDrone2021 [4] dataset, our model showed a 71% reduction in parameter count, an 8.1% increase in mAP50, and a 5.3% increase in mAP50-95 compared to the baseline model. Compared with some excellent detection networks, the model proposed in this paper achieved considerable speed, low parameters, and best accuracy.

CCS Concepts

• **Computing methodologies** → Artificial intelligence; Computer vision; Computer vision problems; Object detection.

Keywords

UAV, Object detection, Small object, High precision, YOLOv8, IOU

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1 Introduction

UAV object detection mainly utilizes computer vision technology and deep learning methods to automatically detect interested objects in UAV images. It has a wide range of applications in agriculture, environmental protection, and urban management. With the continuous development of UAV and optical camera technology, the application scenarios of UAV are becoming more and more extensive, and the object information in UAV images is also becoming more and more abundant, which makes UAV object detection face more challenges.

Compared to conventional object detection, UAV small object detection mainly faces the following challenges. Firstly, small objects may be objects, parts of objects, or specific regions that occupy fewer pixels in the image, making it difficult to extract their features. Secondly, lower pixel values can lead to contrast and occlusion issues, as shown in Figure 1. Therefore, the model must have sufficient complexity to capture the details of small objects.

Scholars have studied the challenges of UAV small object detection and proposed methods like pyramid convolutional neural networks [5], attention mechanisms [6], and multi-scale feature fusion [7]. However, these methods have the disadvantages of large computation and slow speed. In response to the above issues, this paper proposed a high-precision UAV small object detection model based on YOLOv8. This model focuses on achieving precise UAV small object detection while maintaining a lower number of parameters. To achieve this goal, this paper carried out a series of critical designs to improve model performance.

The main contributions of this paper are as follows:

- This paper enhanced YOLOv8 by adding shallow feature fusion layers and new object prediction heads, improving the localization of dense small objects. It also removed the large object prediction head, reducing network parameters and boosting computational efficiency.
- This paper introduced some modules aimed at improving the accuracy of small object detection in the feature fusion network, optimizing the effect of small object detection from several perspectives such as multi-scale, attention mechanism, and feature fusion.
- This paper used a new loss MPDIoU as the localization loss.
- This paper used the LAMP method to perform pruning operations on the trained model.

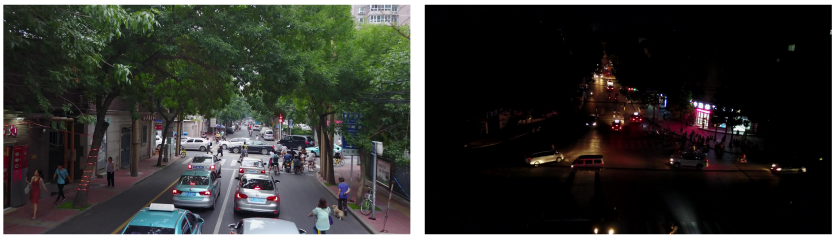


Figure 1: On the left is a small object detection image with occlusion issues and on the right is a small object detection image with contrast issues.

2 RELATED WORK

2.1 Object detection model

Early object detection focused on two-stage algorithms like Mask R-CNN [8], Fast R-CNN [9], and Faster R-CNN [10], which offered high accuracy but lacked real-time performance. This led to a shift towards one-stage algorithms. YOLOv3 [11] was an early example known for its high detection speed and multi-scale predictions. YOLOv4 [11] introduced optimizations like the CSPDarknet53 backbone and PANet to improve accuracy. YOLOv5 [11] followed, offering models of different sizes, new training techniques, and easy-to-use code, making object detection more accessible. YOLOv7 [12] and YOLOv8 further optimized the architecture, enhancing performance on multiple benchmark datasets, and making it suitable for more complex practical application scenarios.

2.2 UAV small object detection model

Small object detection models are a specialized branch of object detection, optimized for small objects in specific scenarios. The SSD [13] (2016) used multi-scale feature maps for different object sizes but struggled with tiny objects compared to multi-stage models like FPN [14]. Faster R-CNN with FPN (2017) improved small object detection but increased computational overhead and was slower than single-stage detectors. Cascade R-CNN [15] enhanced detection accuracy in complex scenes but required more computational resources. EfficientDet [16] (2020) used BiFPN to reduce complexity, which was ideal for resource-limited devices like UAV. Deformable DETR [17] (2021) and Swin Transformer [18] improved detection in complex scenes through advanced architectures but required high computational resources. GCL-YOLO [19] (2023) addressed real-time small object detection in UAV with Ghost-Conv [19] and multi-scale feature fusion, though it faced challenges in complex scenarios and required high-quality datasets.

3 OUR NETWORK

To enhance small object detection, this paper proposed high-precision YOLOv8, as shown in Figure 2.

3.1 Backbone network using Ghost-Conv

UAV images often have wide coverage and high resolution, leading to many low-resolution objects. YOLOv8 was used of 3x3 convolution kernels with a stride of 1, leading to information loss and difficulty capturing subtle features. While a stride of 1 maintains input-output consistency, it increases computational complexity and reduces speed. The small receptive fields of 3x3 kernels are also insufficient for small object detection, which requires larger receptive fields for context. To address these issues, this paper incorporated Ghost-Conv in the backbone feature extraction network.

3.1.1 Ghost-Conv. Ghost-Conv contains redundant features. These features provide additional information in small object detection to enhance the accuracy and stability of the detection. Specifically, small object detection usually refers to the detection of small-sized and proportionally small objects in an image, which may be difficult to detect due to their small size, complex shape, or background interference.

Ghost-Conv's structure, shown in Figure 3, starts with a 1x1 convolution followed by a 5x5 grouped convolution. The resulting features are concatenated with those from the 1x1 convolution. While YOLOv8's 3x3 kernels may cause information loss, Ghost-Conv's 5x5 grouped convolutions better capture small object features, improve the receptive field, and gather more context. Grouped convolutions also reduce computational resources. For embedded devices or weaker hardware, Ghost-Conv's 1x1 Conv layer reduces feature map channels, lowering computational costs and improving network inference speed.

3.1.2 C2f_GhostBottle. This paper proposed a new bottleneck module based on Ghost-Conv. It consists of two Ghost-Convs, and the feature input passes through the two Ghost-Convs to obtain the result, which is then added to the feature input. In the network proposed in this paper, a new feature extraction module C2f_Ghost-Bottom was added, which used a new bottleneck module and replaced the CSPLayer_2Conv module in the original network. The newly introduced bottleneck module in the network not only reduced parameter and computational complexity by compressing the number of channels but also preserved redundant information through residual connection operations. It also increased the depth of the network and improved the model's ability to understand complex scenes.

3.1.3 CADM.. For the down-sampling layer of YOLOv8, directly replacing it with Ghost-Conv will result in a decrease in detection performance. Because Ghost-Conv adopts a grouping strategy, reducing the number of parameters and computation, the richness of feature extraction is insufficient. To address this issue, this paper proposed a new down-sampling module GADM. GADM used an

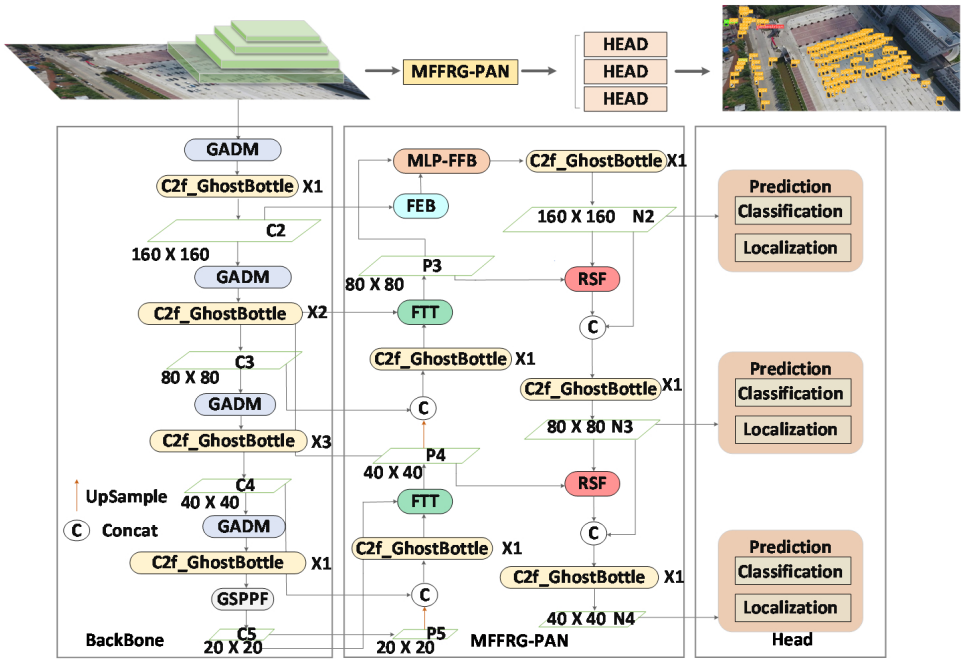


Figure 2: The overall framework of high-precision YOLOv8. C_i represents the feature map of the ith stage from the backbone network, P_i represents the feature map of the ith stage from the PAN [20], and N_i represents the ith stage feature map generated by P_i.

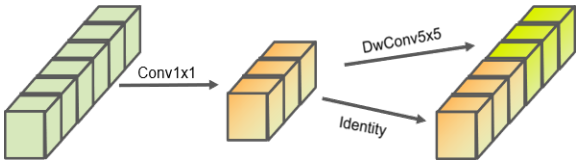


Figure 3: Ghost-Conv.

angular structure to expand the channel dimension to compensate for the shortcomings of feature extraction caused by using grouping strategies. In addition, GADM also used attention mechanisms in the channel and spatial dimensions. The channel attention and spatial attention parts of this module can be combined to provide attention to global context and local features.

3.1.4 GSPPF.. According to Bao, "At the end of the backbone network, the SPPF module has vast calculations due to a considerable number of channels [19]." This paper followed the design concept of Bao and added two Ghost-Convs into the original module SPPF.

3.2 Feature fusion network

In YOLOv8, PAN fuses feature maps from the backbone network, but it has limitations. Small objects in UAV images may suffer from information loss due to fewer pixels and details. PAN's pyramid structure may not effectively capture very small objects due to insufficient depth or density. Down-sampling reduces image resolution, leading to less detailed object information and potential loss of positional data.

In response to the above issues, this paper redesigned a new feature fusion network MFFRG-PAN based on PAN, and proposed three fusion strategies: fusion of C_i and P_j, fusion of P_i and P_j, and fusion of P_i and N_j. The initial feature layer of the backbone network contains rich location information for small objects. This paper enhanced YOLOv8 by adding high-resolution feature fusion in PAN to better utilize this location information. PAN's multi-scale detection heads have varying receptive fields and resolutions. large object heads focus on big objects, while small object heads are tuned for small objects. Removing the large object detection head and adding a new small object detection head improves small object accuracy and reduces parameters and computations, addressing the insufficiencies in YOLOv8's PAN hierarchy.

3.2.1 The fusion strategy of C_i and P_j. Feature fusion from the backbone network and PAN can face challenges, like difficulty detecting occluded objects and feature mismatches. To address this, the MLP-FFB module was designed using a CBAM attention mechanism for high-resolution low-level features. Flattening input features into vectors and applying MLP [21] for learning enhances feature attention better than pooling methods. Directly concatenating low and high-dimensional features could cause mismatches, so appropriately sized convolution kernels were used for different feature scales. This paper used two 7x7 convolution layers in spatial attention to enhance feature dimensions and focus on specific spatial regions, and aid in the handling of occlusion.

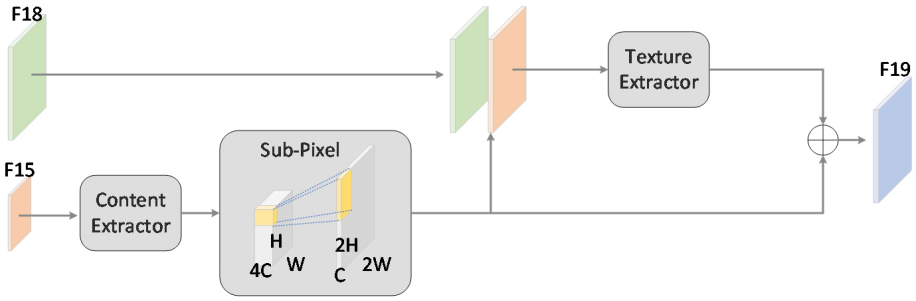


Figure 4: Enhanced version of FTT.

Contextual information is essential for small object detection as it improves position accuracy, reduces false detections, separates objects from backgrounds, and provides size and scale details. To enhance contextual information, this paper introduced the FEB module, which prevented information loss during feature extraction in the YOLOv8 network. Using 3x3 convolutions, dilated convolutions (scale 3), and point-wise addition, the FEB module fuses multi-level features, enhancing the richness and diversity of features, and ultimately boosting the network's performance and expressiveness.

3.2.2 The fusion strategy of P_i and P_j. As shown in Figure 4, F_{15} and F_{18} are the inputs for the enhanced version of FTT. F_{15} is the output of the 15th layer in this paper's network, which is a low-resolution feature map with strong semantics, containing the semantics of objects and scenes and rich contextual information. F_{18} is the output of the 18th layer in this paper's network. As a high-resolution feature map, it provides important local details, object contours, and boundaries. The fine-grained semantic information it provides is essential for detecting multiple objects and distinguishing between different objects in the scene. The enhanced version of FTT builds upon the original FTT, extracting strong semantic and local information from features while filtering out noise.

This paper enhanced FTT's Content Extractor and Texture Extractor by using cross-stage partial connections. The input feature map is divided into two parts: one undergoes convolution, while the other is directly connected to the output through skip connections. This architecture enhanced the information flow within the network. Additionally, the module incorporated weighted residual blocks to more effectively capture features at various scales and levels.

3.2.3 The fusion strategy of P_i and N_j. The dataset also contains slightly larger objects. The RSF module was used to capture detailed information about these objects through spatial attention, thereby improving their accuracy and enhancing the overall object detection performance.

3.3 New IoU loss function: MPDIoU loss

The CIoU loss function used in YOLOv8s is relatively insensitive to numerical changes in aspect ratio, which makes it less effective at measuring differences in object shape and size. Additionally, the use of inverse trigonometric functions in CIoU adds computational complexity. Therefore, this paper introduces MPDIoU as a more effective positional loss function, as illustrated in Figure 5.

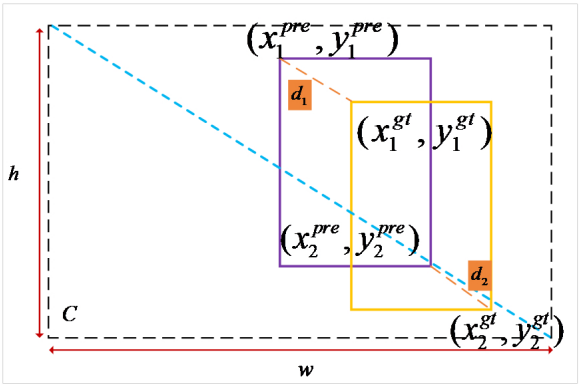


Figure 5: MPDIoU calculation parameter diagram.

MPDIoU uses the upper-left and lower-right points of both the ground truth box and the predicted box to calculate the Euclidean distance between these corresponding points. When the aspect ratio of two prediction boxes with different widths and heights is the same, and their center points coincide with the center point of the ground truth box, CIOU loss may yield the same positional loss. MPDIoU not only avoids this situation, but also considers non-overlapping area, center point distance, width and height deviation, etc. The formula for MPDIoU loss is as follows:

$$L_{MPDIoU} = 1 - IoU + \frac{d_1^2}{w^2 + h^2} + \frac{d_2^2}{w^2 + h^2} \tag{1}$$

Parameter description in the MPDIoU loss function: IoU represents the intersection over union, which is calculated by measuring the ratio between the intersection area of two bounding boxes and their union area. d_1 represents the Euclidean distance between point (x_1^{gt}, y_1^{gt}) and point (x_1^{pre}, y_1^{pre}), and d_2 represents the Euclidean distance between point (x_2^{gt}, y_2^{gt}) and point (x_2^{pre}, y_2^{pre}). w and h represent the width and height of the input image, relatively.

3.4 LAMP

LAMP is a pruning method for deep learning models. First, LAMP reduces the number of model parameters by selectively pruning less important ones during the training process, thereby alleviating

Table 1: Comparison results with the YOLOv8 model measured by mAP50(%).

Models	Pedestrian	People	Bicycle	Car	Van	Truck	Tricycle
YOLOv8n	36.6	29.4	9.0	76.8	40.2	30.0	22.5
YOLOv8s	42.1	33.2	13.3	79.7	45.4	37.5	27.6
YOLOv8m	46.5	36.1	17.8	81.8	49.1	41.0	33.7
YOLOv8l	48.7	38.3	20.0	82.6	49.8	**45.8**	**36.4**
Ours	**54.1**	**43.8**	**20.6**	**85.1**	**50.9**	43.1	35.6

Models	Awning-Tricycle	Bus	Motor	mAP50/%	FPS	Parameter/10^6
YOLOv8n	11.6	48.1	37.5	34.2	154	**3.0**
YOLOv8s	16.2	54.9	44.7	39.5	**156**	11.1
YOLOv8m	17.1	63.5	49.1	43.6	133	25.8
YOLOv8l	19.3	66.0	51.4	45.8	111	43.6
Ours	**20.1**	**67.1**	**55.2**	**47.6**	112	3.2

over-parameterization. Second, LAMP adaptively adjusts the number of parameters in each layer to meet the requirements of specific tasks and datasets. By leveraging pruning techniques, LAMP optimized the model structure based on the importance of each layer's parameters, removed unnecessary complexity, and improved the model's efficiency and adaptability.

The LAMP calculation process is straightforward. For each layer of the neural network, since the weight tensors have different dimensions across layers, LAMP flattens the weight tensors into one-dimensional tensors to unify their dimensions. Afterward, LAMP sorts the tensor weights in ascending order and calculates the LAMP score using the following formula.

$$score\,(u;W) := \frac{(W\,[u])^2}{\sum_{v\geq u}(W\,[v])^2} \tag{2}$$

where u and v are the ascending order numbers of W, weight $W\,[u]$ represents the u-th weight of W, $W\,[v]$ represents the v-th weight of W, score represents the LAMP score of the uth weight of W, and $\sum_{v\geq u}(W[v])^2$ represents the sum of the squares of the weight values whose order numbers are greater than or equal to u.

4 EXPERIMENT RESULT

4.1 Dataset introduction

The VisDrone2021 dataset consists of 6,471 training images, 548 validation images, and 3,190 test images. It is designed for object detection tasks and includes multiple object categories, such as pedestrians, vehicles, bicycles, and motorcycles, among others.

4.2 Comparison experiments with YOLOv8

To further validate the effectiveness of our model, we conducted comparative experiments with YOLOv8 models of different scales, as shown in Table 1.

From the analysis of the experimental results in Table 1

- Table 1 shows that the improved YOLOv8s model achieves an 8.1% increase in overall mAP compared to the original version. AP values for all categories have improved, with notable gains of 10% in key categories such as pedestrians, people, buses, and motorcycles. This demonstrates the enhanced accuracy and overall performance of the improved model in detecting small objects.
- Compared to other YOLOv8 models listed in Table 1, our model demonstrates superior performance in mAP across most categories, achieving the highest average mAP, which is 1.8 points higher than the best result of YOLOv8. Additionally, the parameter size is only 0.2×10^6 units larger than YOLOv8n.
- This is primarily due to the use of Ghost-Conv in the backbone network, which reduced the number of parameters. Additionally, the fusion network incorporated modules specifically designed to enhance small object detection, and the pruning method was applied to reduce complexity.

4.3 Ablation experiments

To explore the influence of various factors in this study, a series of ablation experiments were conducted. These experiments aim to verify the contribution of specific variables to the research outcomes and assess their role within the overall framework. Below is a summary of the main ablation experiments conducted in this study:

- Ablation variable A: Ghost-Conv.
- Ablation variable B: GADM.
- Ablation variable C: added a small object detection layer and removed a large object detection layer.
- Ablation variable D: enhanced FTT.
- Ablation variable E: MLP-FBB+FEB.
- Ablation variable F: RSF.
- Ablation variable G: new IoU loss function MPDIoU.
- Ablation variable H: LAMP.

The following conclusions can be drawn from the Table 2

- In ablation experiment B, GADM used a diamond structure with attention mechanisms to offset Ghost-Conv's impact, improving accuracy. Combined, variables A and B reduced the model's complexity while maintaining accuracy and speed.
- In ablation experiment C, convolutional layers and kernel sizes were adjusted to capture small object features, reducing

Table 2: Results of ablation experiment.

Network	A	B	C	D	E	F	G	H
YOLOv8s								
	√							
	√	√						
	√	√	√					
Ours	√	√	√	√				
	√	√	√	√	√			
	√	√	√	√	√	√		
	√	√	√	√	√	√	√	
	√	√	√	√	√	√	√	√

Network	Params(M)	GFLOPs	FPS	mAP50	mAP50~95
YOLOv8s	11.1	28.8	156	39.5	23.8
	5.7	**18.4**	**166**	38.7	23.1
	5.3	18.8	149	39.7	23.7
	2.9	24.7	123	44.0	26.7
Ours	4.0	28.4	128	44.6	27.6
	4.2	41.2	107	46.5	28.5
	4.3	41.5	111	46.7	28.6
	4.3	41.5	108	46.8	28.8
	3.2	27.3	112	**47.6**	**29.1**

parameters by nearly half. However, the computational cost increased by 5.9 GFLOPs, slowing inference but significantly improving accuracy.

- In ablation experiment D, the enhanced FTT module improved semantic and local feature extraction while reducing noise. Cross-stage partial connections boosted information transmission, further enhancing accuracy.

4.4 Comparative experiments

To evaluate the performance differences between high-precision YOLOv8 and mainstream object detection models, we conducted a series of experiments. This section analyzes the experimental results and presents a comparison chart highlighting the differences in model detection performance. The detailed performance metrics are shown in Table 3.

Table 3 shows that high-precision YOLOv8 ranks third in parameters, has moderate speed, and leads in accuracy among YOLO models. It excels in both speed and accuracy compared to popular models. Table 3 reveals that many lightweight models perform poorly due to reduced parameters, smaller convolution kernels, and shallower networks, which limit feature representation and receptive fields. Lower image resolutions in resource-constrained settings also cause detail loss, complicating small object detection. To enhance lightweight models, consider data augmentation, attention mechanisms, multi-scale detection, model fine-tuning, and improved loss functions.

5 CONCLUSIONS

Many models are currently available for small object detection in UAV, but their accuracy is often limited. This paper optimized the YOLOv8 network by analyzing and redesigning each component, proposing a high-precision YOLOv8 model for small object detection in UAV.

Experiments conducted on the public dataset VisDrone2021 have shown that the model proposed in this paper achieves better performance in the small object detection task compared to YOLO series models and some mainstream models. The model in this paper achieved the best accuracy in experimental detection while maintaining a low number of parameters and moderate computational complexity.

Overall, the model proposed in this paper has achieved encouraging results in UAV small object detection. It is hoped that in the future, with the continuous advancement of technology and the expansion of application scenarios, UAV small object detection model can be further optimized and improved.

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Table 3: results of the comparative experiment.

Network	Params(M)	GFLOPs	FPS	mAP50	mAP50~95
YOLO Series					
YOLOv3-Tiny	8.69	12.9	181	0.164	0.074
YOLOv4-Tiny	6.05	16.5	106	0.165	0.091
YOLOv5n	1.77	4.20	79	0.264	0.133
YOLOv5s	7.04	15.8	86	0.331	0.179
YOLOv5-Lite-G	5.31	15.2	38	0.183	0.086
YOLOv7-Tiny	6.03	13.1	144	0.347	0.180
YOLOX-Tiny [22]	5.04	15.3	111	0.330	0.180
YOLOv8n	3.00	8.7	153	0.333	0.193
YOLOv8s	11.1	28.6	156	0.395	0.238
Popular Models					
Nanodet-Plus-M-1.5x	2.41	**3.4**	59	0.309	0.189
PP-PicoDet-L [23]	5.80	16.8	42	0.379	0.236
GCL-YOLO-S	**1.64**	10.7	44	0.396	0.215
YOLOv10s [24]	8.04	24.5	**185**	0.395	0.239
YOLO-World [25]	12.5	16.0	50	0.457	**0.313**
SuperYolo [26]	4.83	52.1	61	0.433	0.237
Ours					
Ours	3.2	27.3	112	**0.476**	0.291

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